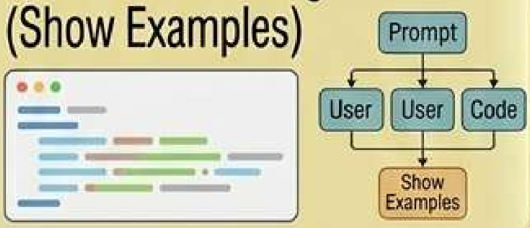
PROMPT ENGINEERING QUICK CHEATSHEET

LAST-MINUTE PLACEMENT EXAM REVISION (1-2 MIN SCAN)

BASIC STRUCTURES & TECHNIQUES

- System, User, & Assistant Prompts
- Clear Instruction vs. Ambiguity
- Define: Goal, Context, & Constraint
- Few-Shot Learning (Show Examples)



KEY TECHNIQUES & SHORTCUTS

- Chain-of-Thought (CoT) ($step1, $step2)

```
steps =
steps1( $step1, $step2)
```

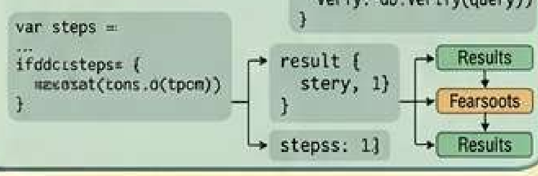
- Check: 'Show your steps'
- Output: table format, code visual'

- Verify: db.**Verify**(query)
- Constraint:

```
steps1 = {
Output: table format, code visual'}
```

- Constraint: 'Max 100 words', 'No repeats'

```
results {
  steps: {..
  verfy: db.Verify(query))
}
```



CONTEXT & DATA MANAGEMENT

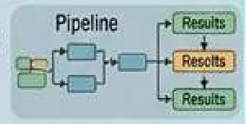
Inject: Document context (rag, database)

- Extract: Key information, named entities

- Clean: Normalize input text
- Check limits: max input tokens

ADVANCED & FLOW

- Tree of Thoughts ($path1, $path2)



- Ask for multiple possibilities
- Verify: check intermediate results
- Consolidate multiple outputs

SAFETY & ETHICS

- Guardrails: Avoid harmful content, bias
- Sanitize inputs for prompt injection
- Implement human-in-the-loop (HITL)
- Verify constraints: 'Do not discuss X'

EVALUATION & ITERATION

- Define metrics: accuracy, coherence, safety

- A/B Test different prompt versions
- Analyze error cases, update context
- Iterate: refine inputs, context, few-shots

ULTRA REVISION ZONE

- CoT: Chain-of-Thought = show steps
- Few-Shot = examples work best
- Constraint: 'max length', 'no repeat'
- Context: Inject necessary data
- Safe: Add guardrails for bias/harm
- Test: Iteratively refine prompts
- Metrics: Define what to evaluate
- Clean: Sanitize inputs & context
- Chain: Connect multiple prompts